

# The Security Implications of Developments in Biotechnology

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# Contents

|                                                           |           |
|-----------------------------------------------------------|-----------|
| <b>Executive Summary</b>                                  | <b>4</b>  |
| <b>Section 1: Advances in Key Biotechnologies</b>         | <b>6</b>  |
| Genomics                                                  | 6         |
| Synthetic Biology                                         | 6         |
| Genome Editing                                            | 7         |
| Convergence with AI                                       | 7         |
| Implications                                              | 7         |
| <b>Section 2: Geopolitics and National-security Risks</b> | <b>9</b>  |
| Biosecurity Threats                                       | 9         |
| The Dual-use Dilemma                                      | 9         |
| Economic Security                                         | 10        |
| Diffusion of Capabilities                                 | 11        |
| Lack of Controls: Ethics                                  | 12        |
| Lack of Controls: Data Protection                         | 13        |
| Broadening Access                                         | 14        |
| <b>Section 3: Looking Ahead</b>                           | <b>15</b> |
| Addressing Weaknesses in Existing Safeguards              | 16        |
| A 'Whole of Society' Approach                             | 19        |
| <b>Notes</b>                                              | <b>20</b> |

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## Cover

*DNA testing in a laboratory (Photo by Westend61 via Getty Images)*

# Executive Summary

This report focuses on the security implications that have arisen as a result of developments in biotechnology in recent decades.

To set the scene, we first look at the rapid advances in genetic-modification technology since the 1970s; the development of synthetic biology since 2000; the research related to genetic engineering, which has brought about technological innovation since 2000; and the innovative aspects of genome-editing technology (CRISPR/Cas9), which first appeared in 2012. We discuss the positive and negative aspects of these technologies, and in particular the importance of advanced life-science technologies from the perspective of risk-benefit analysis and biosecurity.

The ability to understand, recreate and manipulate the basic building blocks of life is driving the growth of the bioeconomy. This includes the commercialisation of biological processes, such as modifying living organisms to perform additional or enhanced functions. Biotechnology is becoming increasingly important for national economic security and competitiveness in global markets; countries are therefore vying for leadership in all of its fields, including its medical, agricultural and environmental applications. The United States, China, the European Union and Japan, among others, have been issuing strategic documents on biotechnology and incorporating it into their economic-security strategies.

Biotechnology has made it possible to manipulate genetic material for medicinal and agricultural purposes, and the opportunities for novel applications have grown exponentially. These ongoing innovations are having a profound impact on health, industry and the environment. But as the manipulation of genetic material becomes more widespread, the risk of accidental release of harmful organisms increases, as does the likelihood of unintended side effects.

The safety and reliability of genome editing are not technically guaranteed, and the ability to manipulate and modify life at the genomic level has led to growing concerns about the impact of editing the human

genome. Such concerns are further heightened by the increasing convergence of biotechnology and artificial intelligence. There are also concerns that the expansion of gene editing could lead to eugenic practices, and about the potential knock-on effects for the offspring of people with edited human genomes. Although no country has yet approved the genome editing of fertilised human embryos, genome editing is currently being tested in laboratories worldwide.

Furthermore, consumer genetic-testing and genome-analysis services are becoming more widely available as providers are increasingly adopting the direct-to-consumer (DTC) model. However, many of these services are often carried out without the involvement of medical specialists, arousing concerns that decisions will be made without a scientific basis, potentially leading to misdiagnoses or inappropriate use. In the US, law-enforcement agencies have also used DTC genomic data to identify suspects in criminal investigations, possibly violating the terms and conditions for the use of personal data. There is an argument, however, that the perceived public good should supersede privacy interests, which would allow courts to accept DTC genomic data in law-enforcement investigations and legal cases.

Though advances in biotechnology create the potential for breakthroughs in medicine, agriculture, environmental management and industry, they also bring risks and ethical dilemmas. In particular, there are concerns that synthetic biology could be used for malicious purposes, such as the production of biological weapons. There is also a risk of accidental release of synthetic organisms, even in controlled laboratory environments. If these organisms spread uncontrollably, they could pose serious threats to public health and the environment. Continued advances in synthetic biology have therefore only fuelled the growing demand for a robust international regulatory framework to manage these risks and prevent misuse.

At present, the multilateral legal framework for preventing biotechnology from having an adverse

impact on international security includes the Biological Weapons Convention (BWC), signed by 187 states, and UN Security Council Resolution 1540. However, within these frameworks there has been no agreement on a verification system to ensure countries are implementing the BWC, or on a mechanism for more in-depth monitoring. Also, because every country has a different stance on biotechnology, based on their culture and ethical values, there is no consensus on what degree of regulation to adopt. It is therefore difficult to formulate globally accepted guidelines. International organisations such as the World Health Organization and European Molecular

Biology Laboratory are advocating careful regulation, transparency, and ongoing public dialogue to encourage the responsible use of genome-editing technology and to ensure it is developed with ethical foresight.

The challenge in regulating emerging technologies lies in the need to constantly expand the scope of governance. In the case of biotechnology, because there are multiple areas of concern – including biosafety, ecology, distributive justice, national security and the impact on climate change – the governance challenge is to incorporate interdisciplinary and multi-sectoral perspectives, and to build a whole-of-society approach.

# 1. Advances in Key Biotechnologies

Developments in biotechnology are progressing rapidly, and there is growing interest in their geopolitical impact. In particular, the development of synthetic biology is expected to make it possible to artificially create various living organisms and materials, and to contribute to the treatment of diseases, the stable supply of food, and the mitigation of climate change through the reduction of carbon-dioxide emissions. The development of genome-editing technology is significantly improving human welfare in various fields, including health, food, climate, energy, and supply-chain resilience.

States will be competing to be the first to develop new biotechnologies, and to use them to gain a strategic advantage. Although biotechnology has the potential to make a profound contribution to collective human prosperity and scientific discovery, the most powerful nations are likely to use their strategic advantage to exploit new biotechnologies, and the markets derived from them, in pursuit of their own geopolitical goals.

Furthermore, synthetic biology and genome-editing technologies could not only have unintentionally destructive effects if their safety is compromised but also could be deliberately misused. The ability to understand and manipulate genetic information could lead to the creation of increasingly effective biological agents and defences against them, potentially sparking a biotechnological arms race.

Over the past two decades, scientists have gained the ability to understand genes, and to manipulate and engineer from scratch the building blocks of life. Overall, these developments have transformed biology from a purely observational science into one where life can be engineered. The following sections provide a brief overview of what is possible with the technology available today.

## Genomics

Genomics refers to the comprehensive study of the structures, functions and interactions of an organism's

complete set of DNA, including those of its genes. This knowledge was greatly advanced by the Human Genome Project (1990–2003), an international effort (although most participating institutions were American, and it was funded partly by the US Institutes of Health) that succeeded in almost completely decoding the human genome sequence by 2003. Subsequent advances in next-generation gene-sequencing technology, along with falling costs, have greatly boosted progress in this field, which has included research that collects and analyses the genomic data of many individuals to elucidate the relationship between cellular physiological functions, genetic traits and disease susceptibility.

## Synthetic Biology

Synthetic biology, an interdisciplinary field combining traditional biology, bioengineering, electrical engineering and information science, involves the redesign of existing natural biological systems to enhance utility or to create wholly new systems. From the outset, synthetic biology has been closely linked to research into infectious diseases, beginning with the complete artificial synthesis of the poliovirus in 2002. Following the creation of horsepox virus in 2017, it is widely believed that the technology to artificially synthesise all viruses has already been achieved.

Furthermore, in 2008, scientists succeeded in synthesising the world's first artificial bacterial genome – a breakthrough in synthetic biology that provided an unprecedented insight into the basic building blocks of life.<sup>1</sup> Efforts to develop and study minimal synthetic cells – those possessing only the machinery necessary for independent life – can also provide a platform for creating custom-designed organisms for various uses, such as drug production, environmental remediation and industrial processes.

Synthetic biology builds upon genomics and rDNA to create and modify biological systems from scratch, pushing the boundaries of what can be achieved in



genetic manipulation. The ability to engineer organisms with novel capabilities could yield unprecedented benefits in health, agriculture and environmental management. However, it could also lead to security challenges such as the creation of new bioweapons.

## Genome Editing

Genome editing has existed in the form of recombinant-DNA technology since the 1970s, but the most significant progress began in the 1990s. In 2012 scientists discovered the CRISPR/Cas9 mechanism and recognised its potential as a genome-editing tool. It spread rapidly around the world due to its accessibility (relatively inexpensive tools for labs) and convenience (it is easy to give it specificity to recognise target DNA base sequences). CRISPR/Cas9 has enabled more scientists to make precise, targeted changes to the DNA of living organisms.

The applications of genome-editing technology using CRISPR/Cas9 are not limited to the rewriting of a single gene but have expanded to include edits that could ultimately affect entire genetic systems of organisms, reshaping their appearance and their responses to their environments.

## Convergence with AI

The convergence of synthetic biology and AI has made it possible to conduct functional-genomics experiments on computers, and research has progressed to the stage where the function of gene sequences can be predicted even before they are assembled, even for combinations that have not yet been seen in nature.

Significant progress is being made in the study of the three-dimensional structure of proteins. These insights have the potential to lead to the development of functional inhibitors and functional stimulants that fit into proteins in three dimensions. Advances in AI technology have enabled scientists to predict the higher-order structure of new proteins with remarkable accuracy, based on the structural data of existing proteins. AI-based protein-structure-prediction technology is expected to provide a significant boost to the pharmaceutical industry, not only because it can accurately predict the structures of effective drugs but also because it enables cost-efficient drug design.

**Along with her co-researcher Jennifer Doudna, Emmanuelle Charpentier was awarded the 2020 Nobel Prize in Chemistry for her work in developing the CRISPR/Cas9 mechanism**



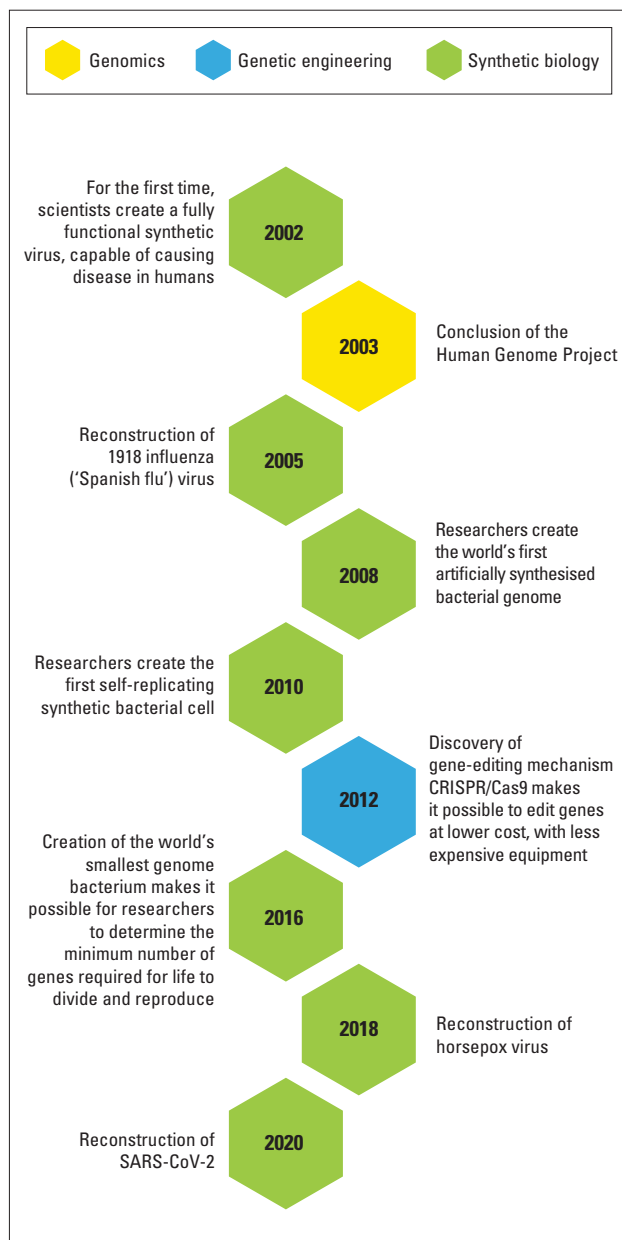
(Photo by Odd Andersen/AFP via Getty Images)

## Implications

Biotechnology is frequently described as ‘dual use’ in the sense that it can be used not only for beneficial scientific purposes but also for harmful ones. One of the main security concerns is that the convergence of emerging technologies could accelerate the identification of harmful genes and DNA sequences, potentially making it significantly easier to turn low-risk pathogens into high-risk threats. This could also facilitate the development of completely new pathogens and even the recreation of extinct high-risk ones.<sup>2</sup> There was an example of this in 2001, when Australian researchers working on a mouse contraceptive as part of a pest-control project unintentionally altered the mousepox virus. This development indicated the possibility of creating a vaccine-resistant variant of a more harmful virus, such as smallpox – a capability potentially applicable to the production of a biological weapon.

Since the development of CRISPR/Cas9 in 2012, these concerns have intensified. Although its application currently requires a certain level of expertise, its use is expected to become easier and more widespread in the near future. Its greater accessibility, in turn, would increase the risk of its misuse – whether intentional or accidental. In addition, the fusion of synthetic biology and AI has created a new security concern in the form of the development of ultra-targeted biological weapons. This issue was highlighted in 2016 by the United States’ then-director of national intelligence, James R. Clapper, in a hearing before the Senate Armed Services Committee on the worldwide threat assessment that

Figure 1.1: **Key advances in biotechnology**



Source: Authors

genome editing could be applied to weapons of mass destruction and their proliferation.<sup>3</sup>

The integration of AI with genetic engineering and synthetic biology could lead to the automation of complex biotechnological processes, which could accelerate their application for beneficial purposes but also increase the pace and scale of their potential misuse. AI could also drive the development of more sophisticated genetic modifications by predicting outcomes or optimising genetic sequences, which raises concerns about the creation of unforeseen or uncontrollable biological systems.

The advances in biotechnology are also leading to the expansion of biomanufacturing, which is creating geopolitical competition and tension. Through biomanufacturing, countries can reduce their dependence on imports of pharmaceuticals, speciality chemical products and speciality materials; and because biomanufacturing can use renewable resources such as plant-derived raw materials, it is often more energy efficient and environmentally friendly than conventional methods. However, biomanufacturing involves advanced biotechnology, and in many cases relies on technological innovations such as synthetic biology, genetic engineering and genetic automation. If intellectual property rights for such technological innovations can be secured, they will potentially boost a country's competitiveness in global markets and enhance its economic security. But there are concerns that biomanufacturing could be misused, for example to produce harmful pathogens. Safety measures, regulatory frameworks and careful monitoring are needed in order to minimise such risks.

## 2. Geopolitics and National-security Risks

While the development of biotechnology could bring great benefits, there are also concerns about its dual-use nature.

### Biosecurity Threats

There are records of biological agents being used as weapons as early as the fourteenth century, but due to their instability and the fact that their effects are often delayed rather than immediate, they have frequently been considered ineffective for modern military use.<sup>4</sup> Biological weapons are completely prohibited by the Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on their Destruction, usually known as the Biological Weapons Convention (BWC), which came into effect in 1975. At that time there was no great concern about the impact of genetic-modification technology on the development of biological weapons, but scientific and technological capabilities have advanced significantly since then.

Public awareness of the threat of bioterrorism has increased in the last 30 years. In 1995 in Japan, it became clear that the Aum Shinrikyo cult had been developing biological weapons such as anthrax and botulinum bacteria; and in 2001 in the US, shortly after the 9/11 terrorist attacks, several addresses were targeted with letters containing anthrax spores, sent through the postal system. These events led to the adoption of UN Security

Council Resolution 1540 in 2004, which requires states to implement effective laws and other measures to prevent non-state actors from acquiring nuclear, chemical or biological weapons and their related materials. In 2005 they also prompted the WHO General Assembly to revise the International Health Regulations to highlight bioterrorism as a key threat, as part of broader efforts to strengthen global health security.

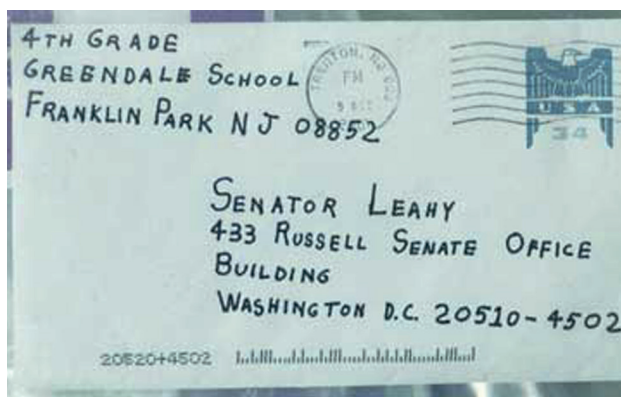
At the same time as the threat of bioterrorism was growing, governments began to recognise the need to strengthen national public-health systems in response to outbreaks of diseases such as Severe Acute Respiratory Syndrome (SARS) in 2003, H1N1 influenza (swine flu) in 2009 and Middle East respiratory syndrome (MERS) in 2012. These outbreaks were followed by the 2014–15 Ebola epidemic in West Africa, which shifted attention from ‘biosecurity’ towards the international control of infectious diseases. Since then, the US has promoted the concept of ‘One Health Security’, in which threats posed by biological agents, whether naturally occurring or man-made, are considered a security issue based on the premise of ‘One Health’, in which human, animal and ecosystem health are regarded as integral parts of a single system.

While in theory the advances in genetic engineering and synthetic biology in recent years could be used to create more virulent or resistant strains of pathogens, or even entirely synthetic biological agents, it is probably more likely that a biosecurity threat will arise from an accidental release, the risk of which has become greater due to the increased manipulation of genetic materials. Indeed, there have already been four incidents in which the SARS virus leaked from laboratories. Two of those incidents were at the same research laboratory in Beijing, resulting in the quarantine of hundreds of people, eight confirmed cases and one death.<sup>5</sup>

### The Dual-use Dilemma

As genomics and synthetic biology inherently possess dual-use characteristics, the challenge lies in promoting

**Senator Patrick Leahy was the intended target of one of the anthrax letters sent through the US postal system in 2001**



(Photo by Getty Images)



legitimate scientific and commercial uses while preventing malicious applications. In response to the anthrax mail incidents of 2001, the US rapidly implemented control measures to closely monitor and control the number of researchers who had access to pathogens and to ensure compliance with research ethics and codes of conduct.

Scientists involved in synthetic biology in particular have been raising concerns about dual use for at least 20 years. A significant milestone was the US National Academies of Sciences' 2004 publication of 'Biotechnology Research in an Age of Terrorism', referred to as the 'Fink report' because of the important role played by the biologist Gerald R. Fink.<sup>6</sup> The Fink report recognised that research into biotechnology carries social responsibility, and underlined the need to maintain a regulatory framework that would entail not only responsible self-management by researchers and research facilities, but also a responsible approach by publishers, who should take into account any potential risks arising from the dissemination of biotechnology research, and by the governments and funding organisations that support the research. Based on the proposals in the Fink report, the National Science Advisory Board for Biosecurity (NSABB) was established in the US as the main national body for examining the potential for 'misuse' of research in the field of infectious diseases. The Fink report remains an important point of reference regarding the dual-use nature of life-sciences technology, particularly research involving infectious pathogens.

Following the Fink report, in 2005 the InterAcademy Panel (since rebranded as the InterAcademy Partnership, IAP), a global network of science, engineering and medical academies, published the IAP Statement on Biosecurity, proposing a code of conduct for scientists that consisted of five aspects: (1) awareness, (2) safety and security, (3) education and information, (4) accountability, and (5) oversight.<sup>7</sup>

As biotechnology advances, many governments, in addition to the US, are recognising its strategic importance. For example, China's strategic biotechnology programme aims to protect the country from potential biological threats and bolster its own technological development.<sup>8</sup> Recognising their dual-use nature, China

**The European Commission is committed to boosting biotechnology and biomanufacturing in the EU**



(Photo by Thierry Monasse/Getty Images)

has also emphasised the importance of developing biotechnologies for military applications, such as for biological defence and enhancing the physical capabilities of soldiers.

In March 2024, the European Commission (EC) published the communication 'Building the Future with Nature: Boosting Biotechnology and Biomanufacturing in the EU', which depicted biotechnology as one of ten technological areas of particular importance to the security of the European economy, due to its 'enabling and transformative nature, the potential risk of civil and military fusion of some biotechnologies, and the risk of their misuse for human rights violations'.<sup>9</sup> For this reason, the EC and the EU member states are carrying out an ongoing evaluation of the security risks associated with biotechnology, including the potential for unauthorised access to these development capabilities.

## Economic Security

Biotechnology will become increasingly important to economic security as it is set to drive a worldwide industrial revolution, with biomanufacturing potentially accounting for one-third of global manufacturing within the next ten years.<sup>10</sup>

The global biotechnology market, which was valued at US\$852.88 billion in 2020, is expected to be worth around US\$3.44 trillion by 2030.<sup>11</sup> The US held a dominant position in the market in 2021, accounting for about 60% of its total value share, followed by the EU (12%) and China (11%).<sup>12</sup> Given the potential benefits of biotechnology across many different sectors, there is a race to apply it in ways that yield strategic advantage. The development of biotechnology will lead to the expansion of

national biomanufacturing industries around the world, and the wide range of innovations expected from the convergence between biotechnology and other emerging technologies will add a degree of complexity and tension to the strategic competition between those states. Several countries – the US, Canada, China, Japan, South Korea, Australia, Saudi Arabia, South Africa and some in Latin America, plus the EU – are formulating national strategies for the bioeconomy, aiming to use biotechnology and biomass to help achieve sustainable economic growth while also accelerating investment in biomedical fields, strengthening agricultural production, and investing in and developing biosafety and quality assurance.

In the US, there was a recent significant development in September 2022 when then-president Joe Biden issued an executive order on Advancing Biotechnology and Biomanufacturing Innovation for a Sustainable, Safe, and Secure American Bioeconomy.<sup>13</sup> This represented an important step forward towards a whole-of-government approach aimed at leveraging biotechnology and biomanufacturing to boost the US economy and strengthen national security, and can be seen as an attempt to counter China's growing capabilities in this field. The specific objectives of the executive order included strengthening the United States' competitiveness in biotechnology by investing in research and innovation, as well as creating jobs in the sector.

In March 2023, the Office of Science and Technology Policy published the report 'Bold Goals for U.S. Biotechnology and Biomanufacturing', outlining six targets for the implementation of the 2022 executive order: data protection, infrastructure development for biomanufacturing, development of human resources, regulatory clarity and efficiency, biosafety and biosecurity, and international collaboration in research and development.<sup>14</sup> Achieving all of these goals and requirements will indeed require a whole-of-government approach, as well as public-private partnership efforts.

The EU has also adopted various strategies to foster the development of biotechnology, aiming to ensure that member states remain competitive in this area while also addressing ethical, safety and regulatory concerns. The European Parliament approved the first European bioeconomy strategy in 2012, with a document entitled 'Innovating for Sustainable Growth: A

Bioeconomy for Europe' that emphasised investment in research, innovation and skills.<sup>15</sup> Updated in 2018, the strategy document now reflects the EU's climate and biodiversity-conservation commitments, and emphasises sustainable biomass production, biodiversity protection and bio-based innovation.<sup>16</sup> By integrating biotechnology innovation into these strategies, the EU is ensuring that economic growth and technological progress contribute to environmental protection and the sustainable use of natural resources.

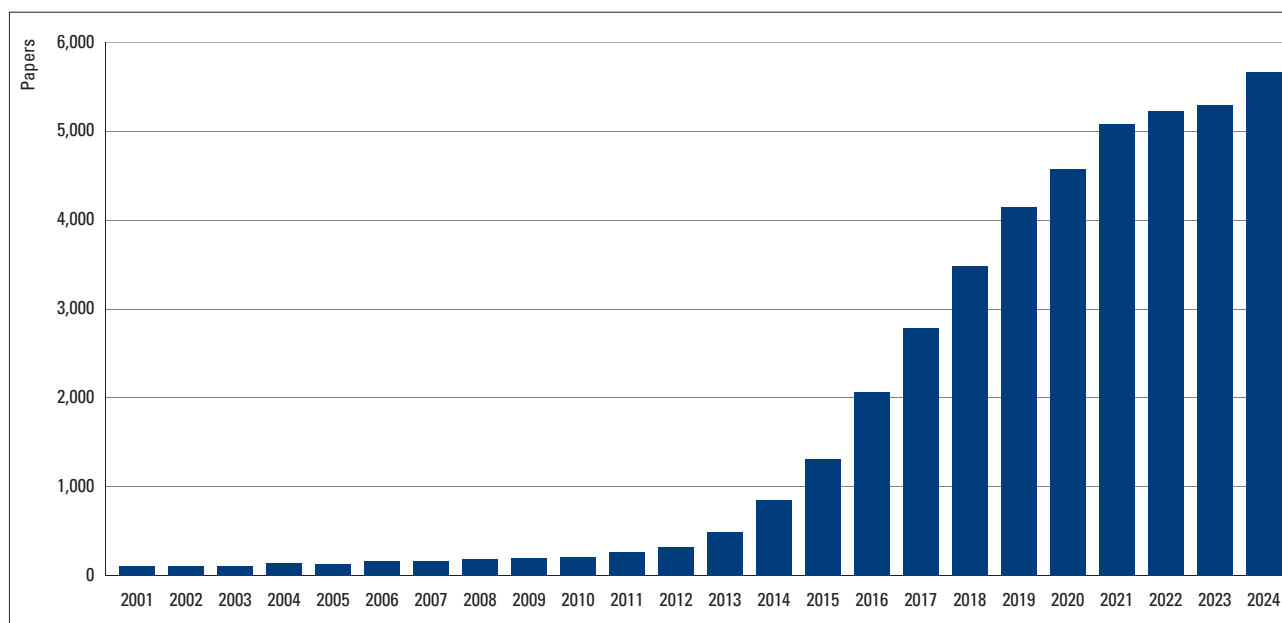
Japan, meanwhile, drew up its first bioeconomy strategy in 2019, setting the goal of becoming 'the world's most advanced bioeconomy society' by 2030.<sup>17</sup> The key elements of the strategy were sustainability, a recycling-oriented society, and improved public health through the wider use of biotechnology. More recently, in June 2024, Japan formulated an updated bioeconomy strategy that reflected the impacts of the COVID-19 pandemic and the changing international situation in Ukraine and the Middle East; the pursuit of carbon neutrality; the circular economy, including supply-chain disruption and trade disputes; and the need to guarantee food and energy security.<sup>18</sup> The updated strategy seeks to address the issues of Japan's decreasing and ageing population, and the country's heavy reliance on imports of food, energy, pharmaceuticals and several other goods, while presenting a pathway to growth through investment in science and innovation, including in the biotechnology field.

## Diffusion of Capabilities

The discovery in 2012 of CRISPR/Cas9, a versatile gene-editing mechanism, has made it possible for laboratories to edit almost any gene without requiring additional specialised technology. With its lower cost and relative accessibility, the CRISPR/Cas9 mechanism has spread rapidly, catalysing a tenfold expansion in gene editing worldwide since its discovery.<sup>19</sup>

Post-CRISPR/Cas9, genome editing has become even more sophisticated. The latest CRISPR/Cas9-based technologies can not only alter genomic DNA but also the components that may affect gene expression, and the range of applications is expanding. Other tools for genome editing, not based on CRISPR/Cas9 technology, have also developed in parallel.

Figure 2.1: **Number of papers with the keywords ‘genome editing’ in the PubMed\* database**



\*PubMed is a free database containing more than 37 million citations and abstracts of biomedical and life-sciences literature.  
Source: Authors

Achieving meaningful results using genome-editing tools currently requires expertise in molecular biology and genetics, bioinformatics skills, and sophisticated laboratory techniques. At the same time, data analysis and interpretation are also needed, involving biostatistics and advanced data-processing techniques. That is why the most significant and reproducible results are often achieved by interdisciplinary teams that combine these skills.

However, the availability of genome-editing tools is expected to increase significantly in the near future, making it possible to achieve meaningful results with less specialised knowledge. New bioinformatics tools are already appearing that can be used effectively by people with little bioinformatics training. In addition, the falling costs of genome editing are expected to make it possible for small-scale laboratories, start-up companies and universities to conduct genome-editing research and applications that could previously be undertaken only by institutions with greater financial resources. Currently, strict regulations – Biosafety Levels 3 and 4, as defined by the World Health Organization – are imposed on laboratories that handle pathogens deemed to be highly infectious. So, in theory, there is some level of oversight of access to higher-risk pathogens. However, as genome-editing tools become

more accessible, clearer guidelines and policies are needed, particularly for non-specialists and in less regulated environments.

## Lack of Controls: Ethics

The ability to engineer or manipulate life at the genomic level has given rise to concerns about the impact of editing the genomes of humans, where the safety and reliability of genome editing has not been technically guaranteed. While this technology could be used to intervene in fertilised human embryos to avoid severe genetic diseases such as muscular dystrophy or Huntington’s disease, there is not yet sufficient evidence that such procedures will not have adverse effects such as the activation of cancer genes or the knockout of tumour-suppressor genes.<sup>20</sup> There are also concerns that the expansion of gene editing could lead to eugenics, and could have complex sociological implications for those future generations impacted by edited human genomes.

Recognising the ethical implications of genome editing, the first International Summit on Human Genome Editing convened prominent molecular biologists and ethicists in Washington DC in 2015.<sup>21</sup> However, the technological advances continued to be applied, and in 2018 a Chinese researcher, He Jiankui, controversially

### Chinese scientist He Jiankui at the Second International Summit on Human Genome Editing in 2018



(Photo by S.C. Leung/SOPA Images/LightRocket via Getty Images)

announced at the Second International Summit on Human Genome Editing that he had edited the genomes of fertilised human embryos to prevent HIV. This was seen as crossing a line, and in 2019 he was convicted for violating medical regulations and imprisoned by the Chinese government.<sup>22</sup>

Genome editing of fertilised human embryos has not been approved anywhere. In 2020, a survey covering 106 countries showed that 96 of them had some form of policy document (a law, regulation, guideline, code or international treaty) relating to the use of genome editing to modify early-stage human embryos, gametes or their precursor cells.<sup>23</sup> However, on the premise that the embryo will not be returned to the uterus and will not subsequently develop into a human, the process is currently being carried out at the research level in many countries.

If the current trajectory of technological development continues, it may be possible to use genome editing in clinical applications (for example, uterus transplants of genome-edited babies) in the near future under the premise of 'treating serious genetic diseases' rather than pursuing any eugenic aim. While there is a need for thorough discussion of the ethical aspects, there are many medical professionals and pharmaceutical companies who want to use this technology to develop medical care. Given the potential to alleviate human suffering from genetic diseases, it may prove difficult for governments to maintain the argument for postponing the application of genetic editing to clinical treatment.

## Lack of Controls: Data Protection

Consumer genetic-testing and genome-analysis services are also expanding through the direct-to-consumer (DTC) model, where the leading companies include 23andMe, DNA4Life, Illumina and DNA Fit. The global consumer genetic-testing market was estimated to be worth almost US\$2bn in 2023 and is expected to be worth more than US\$8bn by 2033.<sup>24</sup> There are limited controls on these consumer genetic-testing services, and on how their diagnoses are used. Often the tests are carried out without the involvement of a medical professional, so there is a risk that decisions will be made without a scientific basis. And if the initial test results are only representative of a specific group of people, or if the sample size is not large enough, it will be necessary to carry out further studies to ascertain whether the initial results can be replicated.

In the US there are also cases where law-enforcement agencies have used DTC genomic data to identify suspects in criminal investigations. In such cases it is possible that there have been violations of the terms and conditions for use of personal data (for example, that it must be used only for the purpose for which it was originally collected). There is an argument, however, that the perceived public good should supersede privacy interests, which would allow courts to accept DTC genomic data in law-enforcement investigations and legal cases.

The question also arises of what happens to the DNA information of customers if the DTCs involved in genetic testing are acquired by other companies. This possibility was explored in November 2024 in the context of the financial difficulties experienced by 23andMe, which possesses a vast and highly valuable DNA database.<sup>25</sup>

DTC genetic testing is already regulated by law in many developed countries, but in some others, such as Japan, it remains subject only to guidelines. As single nucleotide polymorphism (SNP) testing and sequencing analysis of the human genome has become available in many research facilities and companies, the testing work itself tends to be outsourced to low-cost companies, a significant proportion of which are located in China.<sup>26</sup> Against this backdrop, there are suspicions that the genome-sequencing data of humans, animals and plants is not being properly managed, and that



**A direct-to-consumer DNA-testing kit produced by the company 23andMe**



(Photo by Tiffany Hagler-Geard/Bloomberg via Getty Images)

information obtained through contract analysis is being accumulated in China. The Biosecurity Law of the People's Republic of China, which came into effect in 2021, has a strong focus on promoting research and development in biotechnology and on strengthening the country's biosecurity, but does not adequately address ethical considerations or measures to ensure transparency and a more equal global distribution of the benefits of biotechnology.<sup>27</sup>

That said, China has strict laws in place to regulate the collection and export of genetic data, including its 2019 Administrative Regulations on Human Genetic Resources.<sup>28</sup> These laws require approval for the collection of genetic data and restrict sharing with foreign entities. China's Personal Information Protection Law, enacted in 2021, also includes strong protections for personal data

and is similar to the General Data Protection Regulation of the EU, which is one of the world's strictest data-protection laws and applies to DTC genetic-testing companies operating in EU countries.<sup>29</sup>

## Broadening Access

The COVID-19 pandemic exposed issues such as the vaccine gap and the limited supply of therapeutic drugs, which were largely due to delays within the international information-sharing system, the disparity in technological capabilities between countries, and the fact that the provision of medical care relies on countries' political stability and sanitary environment. The pandemic showed the importance of sharing accurate information internationally, of anticipating crises as early as possible in order to take measures to contain highly infectious diseases, and of political cooperation and mutual support between affected countries.

It is possible that the benefits of biotechnology will be distributed very unequally, reinforcing existing health and economic disparities both between and within countries. To increase the chances of the benefits of advances in genomic technology becoming much more widely available, coordinated international action on issues such as the privacy of genetic information, consent and the potential for genetic discrimination will be needed. And to deter the misuse of biotechnology in ways that threaten individual privacy and human rights, robust governance systems will also be necessary.



### 3. Looking Ahead

With the rapid development of biotechnology that is set to take place in the next couple of decades, we can expect groundbreaking developments to reshape healthcare, agriculture, environmental management and even human evolution. In medicine, improvements in genetic diagnostic technology (polygenic risk scores, for example) will allow for more accurate prediction of disease occurrence, leading to advances in preventive medicine. The development of drugs to treat cancers, metabolic disorders and neurodegenerative diseases will accelerate. The world's three major infectious diseases – malaria, tuberculosis and AIDS – should be largely overcome, or at least the number of people affected will decrease significantly.

Genome editing may also be used to manipulate fertilised human embryos with the aim of combatting severe genetic disorders, opening the way for its integration into medical treatments and applications. However, the debate on manipulating genomes for the enhancement of certain traits will continue, probably with an increasing focus on what restrictions should be imposed on medical applications.

In agriculture, genome editing and other applications of biotechnology will enable the efficient growth and preservation of high-nutrition crops. Environmentally friendly energy policies, such as the development of biofuels, will progress. However, it is unclear whether global warming can be significantly slowed.

Synthetic biology and bioengineering will continue to progress, and scientists will be able to design completely new organisms with specific functions (such as bacteria that produce biofuels), modify organisms, and synthesise pharmaceuticals. Synthetic biology allows for more targeted treatment of previously untreatable diseases, such as certain cancers and genetic disorders, with personalised drugs and cellular therapies tailored to an individual's genetic profile. On the other hand, the ability to modify human cells raises ethical concerns about human enhancement, leading to debates about 'designer babies' and human genetic engineering.

Synthetic biology is currently in the research stage, with examples of its capabilities being reported in leading scientific journals. In future it is likely to contribute to more accurate medical diagnoses, and to more effective and less expensive treatments.

Releasing genetically engineered microbes into the environment could also break down pollutants, clean up oil spills, degrade plastic waste, and make the air, soil and water cleaner. In addition, designing organisms that absorb more carbon dioxide, and producing biofuels to replace fossil fuels, could help combat climate change. On the other hand, the release of genetically engineered organisms into natural ecosystems could have secondary ecological consequences, such as spreading engineered traits to wildlife and unintentionally altering ecosystems.

In agriculture, crops genetically engineered to increase productivity, pest resistance and climate-change tolerance could help feed the rapidly growing global population, especially in regions facing food insecurity due to climate change. However, as is already the case today, genetically modified organisms (GMOs) may face resistance from the general public, and there may be growing concerns over the monopolisation of agricultural biotechnologies by large corporations.

Above all, synthetic biology raises concerns about the misuse of GMOs for the manufacture of bioweapons and for malicious purposes such as bioterrorism, as the ability to custom design pathogens makes it easier for malicious actors to create harmful biological agents. But even in controlled laboratory environments there is a risk of accidental release of synthetic organisms. If these organisms spread in an uncontrolled way, they can pose serious public-health and environmental risks. Therefore, as synthetic biology becomes more potent, there is a growing need for a robust international regulatory framework to manage risks and prevent abuse.

Synthetic biology has the potential to revolutionise healthcare, agriculture, environmental management and industry in the next 20 years or so, but in order for

it to benefit society as a whole, a balance must be struck between the need for innovation on one hand and legitimate concerns about safety, fairness and ethics on the other. The US, China, several European countries and some elsewhere, such as Australia, South Africa and South Korea, are already investing heavily in the bioeconomy. The development of biotechnology innovations (such as gene editing, synthetic biology and precision medicine) will be one of the key elements in determining a country's level of competitiveness. Countries will continue to compete to attract the best talent in biotechnology, secure intellectual property rights and build cutting-edge biotechnology industries.

Biotechnology is also closely linked to national security, including economic security. The ability to rapidly develop vaccines, counter bioterrorism and protect against pandemics confers a strategic advantage. Competition will also escalate over gene-editing technologies such as CRISPR and advances in bioengineering that could lead to military or dual-use technologies. Genetic data is likely to become a strategic resource, as countries with access to large genetic data sets will be able to advance research in precision medicine, gene therapy and population-level health monitoring. Privacy and data-security concerns are likely to lead to international competition over the collection, management and ethical use of genetic information.

It is unclear how some countries will fare in this competitive environment. In Japan, for example, there are excellent research capabilities and government support in the biotechnology domain, but Japanese companies face significant challenges when competing with the world leaders, which are mostly in the US, the UK, France, Germany and Switzerland, with China also on the rise. The reasons for this include the lower pace of innovation in Japan, lower risk tolerance in the investment culture, and issues related to the decreasing and ageing population. If tensions between major world powers increase, there is also a possibility of 'biotechnology decoupling', in which the supply chains for important biological products such as vaccines, antibiotics and genetically modified crops are divided into separate blocks.

The other side of the coin is that international cooperation in the field of biotechnology is also expected to

develop. The COVID-19 pandemic has shown the importance of international cooperation in vaccine development and distribution. Biotechnology cooperation in the context of pandemics, infectious diseases and drug-resistant bacteria can become a major area of partnership between countries, especially through global health organisations such as the WHO and Gavi, the Vaccine Alliance. There is also potential for countries to work together to build more resilient global supply chains for important biotechnology products such as vaccines, diagnostic tools and therapeutic procedures. Multilateral efforts to ensure equitable distribution of these products could prevent competition from escalating into conflict.

Multilateral research initiatives in areas such as gene therapy, cancer treatments and climate-resilient agriculture could also promote international cooperation – as did the Human Genome Project, which brought together scientists from multiple countries. Above all, biotechnology is key to addressing global challenges such as climate change and food security, so it could be hugely beneficial if countries work together on biotechnology-driven solutions such as climate-resilient crops and bioengineering algae to absorb carbon dioxide.

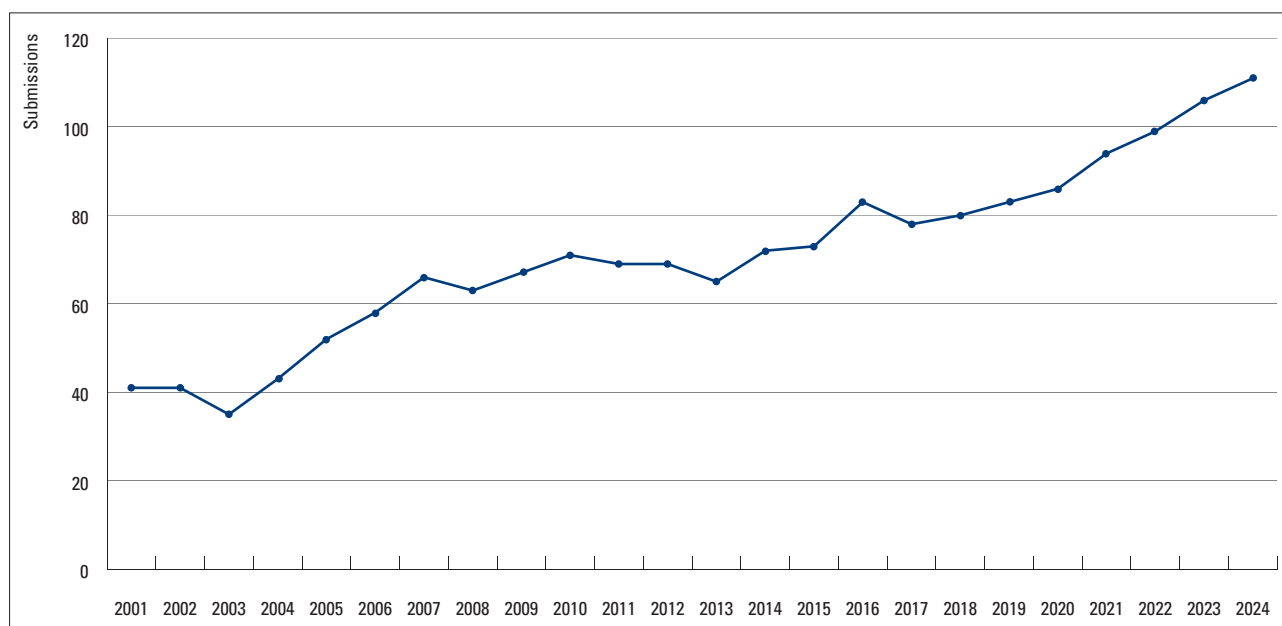
## Addressing Weaknesses in Existing Safeguards

The rapid advances in biotechnologies often outpace the relevant regulatory frameworks, increasing the risk of unintended consequences or deliberate misuse. It is therefore necessary to enhance the global governance frameworks to monitor and control the use of these technologies.

### The Biological Weapons Convention

The BWC is a multilateral framework that prohibits the use of biological agents as weapons. It bans biological agents and toxins 'of types and in quantities that have no justification for prophylactic, protective or other peaceful purposes'.<sup>30</sup> Its Article X requires that the parties to the convention cooperate for 'peaceful purposes', although the precise criteria for those are not defined, and stipulates that the convention should be implemented in a way that does not hamper the economic or technological development of states parties 'in the field of peaceful bacteriological (biological) activities'.

Figure 3.1: **Number of CBM reports submitted to the United Nations Office for Disarmament Affairs (UNODA) by states parties to the BWC**



Source: UNODA

However, the BWC does not provide verification measures to confirm states parties' compliance. There are political reasons for this. The treaty was negotiated during the Cold War between the US and the Soviet Union, and verification measures – including intrusive on-site inspections – were not acceptable at the time.<sup>31</sup> Indeed, with no verification system in place, it has been revealed that in the mid-1970s the Soviet Union was formulating a large-scale secret biological-weapons programme, the *Biopreparat*, that used civilian research as a cover and included work on weaponising pathogens such as anthrax, smallpox, bubonic plague and tularemia.

Attempts were made during the period 1996–2001 to establish a verification system to ensure compliance with the BWC, but no agreement was reached. Since then, there have been further attempts to strengthen the convention, such as through confidence-building measures (CBMs), mechanisms for international cooperation and assistance, and mechanisms for reviewing scientific and technological progress – but as none of these are legally binding, their effectiveness is limited. Particularly since 2001, when the negotiation of a legally binding verification system was halted, the utility of CBMs has been emphasised as a means of at least increasing the transparency of BWC implementation by states parties. The number of countries submitting

CBM reports (see Figure 3.1) has been increasing over the years, but because there is no legal obligation to do so, the submission rate in 2023, the year with the highest number of submissions, was only 57% (105 out of 185 countries). Also, the contents of the CBM report vary from country to country, making a standardised evaluation impossible, so the submission of the CBM report is not in itself an indication of the effectiveness of BWC implementation.

At present, neither the states parties to the BWC nor the United Nations, which supports the implementation of the BWC, are responding adequately to the development of biotechnology. The stakeholders have not agreed, for example, on the fundamental question of whether the development of biotechnology should be addressed as a security issue to be regulated within the framework of disarmament and non-proliferation.

Meanwhile, the three BWC Review Conferences that have taken place since 2011 have addressed the issue of how to respond to research that raises dual-use concerns (DURC). Scientific and technological developments with dual-use potential require continuous checks and monitoring by experts, and the question of who should be conducting such monitoring is being discussed at the BWC, partly because there is no certainty about whether the scientists of the member states

are truly independent of the political motivations of their governments.

In addition to the review of scientific and technological progress, there has also been a long-standing parallel discussion about the importance of education, awareness and norms for scientists and researchers directly involved in biotechnology research and development that has dual-use potential. Within research institutions, an increasing number of awareness surveys have been conducted regarding the prohibition of biological weapons and the non-proliferation of weapons of mass destruction. In 2021, Tianjin University in China, Johns Hopkins University in the US and the InterAcademy Partnership (IAP) jointly created 'The Tianjin Biosecurity Guidelines for Codes of Conduct for Scientists', and efforts are being made to disseminate these guidelines via the BWC Review Conferences.<sup>32</sup> The content of the guidelines is not political, but instead presents principles supported by the IAP. Overall, this is a notable example of a context where China and the US can cooperate, at least at a peer-to-peer level.

### UNSC Resolution 1540

While the BWC primarily focuses on prohibiting states from using biological weapons, there is also UN Security Council Resolution 1540, which was adopted by the UN Security Council in 2004 as a framework for preventing the proliferation of weapons of mass destruction, with a particular focus on the dangers of such weapons falling into the hands of non-state actors.<sup>33</sup> Resolution 1540 is the most universally applicable framework that addresses the dangers of biological weapons, as it obliges all UN member states to implement it under Chapter VII of the UN Charter. Resolution 1540 serves to reinforce the BWC by requiring all UN member states to implement it, with implementation subsequently reviewed by the 1540 Committee, a subsidiary body of the Security Council. However, Resolution 1540 does not define what constitutes a biological weapon, nor does it indicate a common understanding of what kind of legal framework is needed to prevent the proliferation of biological weapons to non-state actors. It only obliges UN member states to develop legislation to prevent the proliferation of weapons of mass destruction and their means of delivery to non-state actors, leaving

### The UN Security Council unanimously adopts Resolution 2663 in November 2022



(Photo by Loey Felipe via UN Photo)

it up to the member states themselves to decide how to do this. By not making a judgement on what should be considered a biological weapon and controlled, and how, Resolution 1540 leaves countries free to proceed on the basis of their own level of scientific and technological development.

Nevertheless, Resolution 1540 provides a useful framework for dealing with the dual-use issues arising from recent developments in synthetic biology. In addition to ensuring the effective control of biological substances, it obliges countries to control flows of funds associated with biotechnologies that raise dual-use concerns, and enables more effective responses to those concerns.

Given the pace of biotechnological developments, the traditional methods aimed at safely controlling micro-organisms are no longer sufficient, and it is becoming vital to ensure that genetic information and other data are managed appropriately. Current information management in the biotechnology field includes publishing research results in academic journals, presenting at academic conferences, sharing information via the internet, and sharing information among researchers working in the same laboratories. But as many cutting-edge research projects become more international and interdisciplinary, and as the sources of funding and investment become more diverse, the importance of controlling the intangible transfer of technology and information is being increasingly emphasised.

A comprehensive review of the national implementation of Resolution 1540 by UN member states, compiled in 2022, pointed out that the rapid development of science and technology and the expansion of global

commerce have increased the risk of misuse or abuse of the products resulting from technological progress.<sup>34</sup> Following the comprehensive review, Security Council Resolution 2663, adopted in November 2022, encouraged UN member states to take measures to control the transfer of intangible technology and information that have the potential to be used for weapons of mass destruction and their means of delivery.<sup>35</sup>

## **A 'Whole of Society' Approach**

There is widespread awareness that the possible applications of biotechnology give rise to ethical issues that require a whole-of-society approach – a point made clearly, for example, at an Organisation for Economic Cooperation and Development (OECD) conference in 2021.<sup>36</sup> Such an approach would recognise that technological innovation needs to be combined with inclusive and responsible governance structures, taking into account ethical, social and environmental factors. In particular, the OECD conference advocated the creation of a framework conducive to transparency, social responsibility and fairness in the development of emerging technologies that have significant social impact, such as AI and biotechnology. The conference stressed that a whole-of-society approach requires both

policy and social readiness, and that collaboration is needed at the local, national and international levels to promote transformative change in a sustainable and socially responsible way.

Internationally, there is no consensus on regulating biotechnologies, as each country adopts its own stance based on its cultural and ethical outlook. But in order to prevent the misuse of such technologies while promoting their peaceful application, it is essential to strengthen international governance. This includes the need for regulatory frameworks to prevent the military diversion of technologies, and ensuring international consistency in biosafety standards. It is also important to establish a system in which researchers, governments and the private sector work together and share ethical responsibilities.

When dealing with emerging technologies, expanding the scope of governance is always a challenge. Nowhere is this truer than in the case of biotechnology, which, as we have seen, has wide-ranging security implications. It does indeed appear to be the case that the best way to tackle the problems it poses, and indeed to grasp the benefits it offers, is to pursue an approach that is multidisciplinary and includes all sections of society.



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